Aaryan Sharif

London, UK · aaryan.sharif24@imperial.ac.uk · +44 7917 979050 · linkedin.com/in/aaryansharif · github.com/aary1n

EDUCATION

Imperial College London – MEng Materials Science & Engineering Sep 2024 – Jun 2028 (expected)
Current average: 74.6% – First Class · Focus: Applied Mathematics & Computational Modelling
Relevant mathematics: Probability & Statistics – **100%** component · PDEs · Tensor & Vector Calculus

ACHIEVEMENTS

- **QuantiHack** – sponsored by Jane Street, Optiver & Anthropic: ranked **1st/800** by cumulative PnL across five trading days after identifying execution throughput as the bottleneck and parallelising order placement; qualified for the London finals.
- **Full-Tuition Academic Scholarship**, Imperial Department of Materials – merit award.
- **Robert Warner Bursary** (sole cohort recipient) · **Imperial College UROP Bursary** (1 of 60 college-wide) · **Department of Materials UROP Funding** (awarded twice) · **upReach Tech500 Associate**.

QUANTITATIVE RESEARCH

Chipper – Independent Market Microstructure Research Feb 2026 – Present

- Built a Python/Polars capture engine that recorded and sequence-validated a ∼**25 GB** limit-order-book dataset from a live WebSocket feed (historical L2 depth is not published by the exchange).
- Engineered a 10-feature microstructure set spanning multi-depth book imbalance, microprice, spread, order-flow imbalance, realised volatility and liquidity withdrawal; trained a logistic-regression baseline for 1 s microprice direction under segment-respecting walk-forward CV with a one-day embargo: out-of-sample **AUC 0.579 ± 0.005**; fee-aware replay showed the edge did not clear 20 bps round-trip taker costs.

RESEARCH EXPERIENCE

Contractor, Anthropic Sep 2026 – Present

- Evaluating frontier-model behaviour for AI safety: built a custom test harness to probe edge cases and wrote reproducible failure reports that engineers acted on.

Research Intern (UROP), Imperial College London – Oxborrow Group Jun 2026 – Present
Room-temperature maser modelling · first-author manuscript in preparation

- Built and validated a Python-driven COMSOL eigenmode model of a room-temperature maser resonator (∼1.45 GHz), reproducing two independent published datasets before extending it to new cavity designs.
- Resolved a ∼30% discrepancy between two published quality-factor measurements by separating dielectric and conductor-wall losses.

Universal Model for Atoms, Meta FAIR – Preview Benchmarking Jul 2025 – Sep 2025

- Selected for gated preview access; benchmarked UMA checkpoints against MACE and NEP interatomic potentials in PyTorch/ASE – force/energy convergence and failure modes; automated the HPC evaluation across 50+ configurations, cutting per-structure setup from ∼15 min to < 1 min.

Research Intern (UROP), Imperial College London – Wenman Group Jul 2025 – Aug 2025
Atomistic modelling of Zr/ZrO_2 interfaces

- Deployed the NEP/GPUMD machine-learning potential stack to model Zr/ZrO_2 interfaces relevant to oxide growth and nuclear-fuel cladding degradation; built and open-sourced `zrbuildr` for parameterised slab/interface generation and lattice-matched supercell search.

SELECTED PROJECTS

Grapple – Real-Time ML Under a Hard Latency Budget Nov 2025 – Present

- Wrote the C#/.NET host: Python inference workers share frames via memory-mapped arenas with a FlatBuffers schema; a lock-free 120 Hz cursor loop extrapolates between ∼15 Hz tracking updates.
- Built the low-latency cursor path of a real-time gesture-control runtime: PyTorch-trained MobileNetV3 served via ONNX Runtime against a 5 ms latency target – INT8 **P50 0.83 ms / P95 1.61 ms**, ∼50% lower latency than FP32 under the same benchmark protocol.
- Designed sensitivity-guided static INT8 quantisation (27/53 convs, per-channel): naive full-graph INT8 inflated cursor-tracking MSE by 56%; probing conv sensitivity on a split disjoint from calibration and eval cut the degradation to **+1.4%**.

SiPhON – C++/Python Numerical Simulation Library Jan 2026 – Present

- Implemented a C++/pybind11 shift-invert Arnoldi eigensolver (Eigen/Spectra) with zero-copy NumPy bindings and Monte Carlo propagation of fabrication tolerances to eigenvalue uncertainty.

TECHNICAL SKILLS

Languages: Python · C++20 · C#/.NET
Libraries & Tools: NumPy/SciPy · Polars · pandas · PyTorch · Linux · Git · PBS/HPC